

WEEKLY PROJECT PROGRESS REPORT (WPPR)–

For week commencing 7 March 2026

Programme: BCA (Artificial intelligence, Cloud Computing & Devops)

Student Name: Mohamed rifan p

WPPR: WEEK 3 Register Number: 23BCAICD052 Internal Guide's Name: MISS Niharika

MAJOR PROJECT Title:

Cloud Cost Intelligence for Retail

Targets set for the week:

The targets for week three were to implement the Anomaly Investigator agent with a 14-day rolling Z-score, add a configurable Z-score threshold slider (range 2.0 to 5.0) to allow users to adjust anomaly sensitivity, develop the Spend Driver agent to compute month-over-month variance by service type and retail department, add a monthly spend trend line chart to the Overview tab, add a cost-by-service bar chart, and performance test the system with 10,000 rows.

Progress/Achievements for the week:

All targets were achieved. The <code>rolling_zscore()</code> function was implemented and successfully flagged anomalous days where daily spend deviated significantly from the rolling baseline. A Z-score slider was added to the sidebar, providing real-time adjustment of the anomaly threshold. The Spend Driver agent was built using the <code>variance_drivers()</code> function, which identifies the top eight contributors to month-over-month cost changes, along with their impact percentages. A monthly spend line chart and a cost-by-service bar chart were added to the Overview tab, giving users clear visual trends. Performance testing with 10,000 rows showed a total processing time of 1.6 seconds – well within the target of 5 seconds

30-03-2026

Future Work Plans:

The next week will focus on completing the remaining two agents: Unit Economics (cost per usage unit by service) and Optimization Scout (idle resource heuristic). Additionally, regional cost breakdown and unit-cost bar charts will be added to the Overview tab. The synthetic dataset generator will be extended to support up to 50,000 rows

Implementa on shown: No

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Yes

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Remarks by the Internal Guide:

Signature of the student

Signature of the Internal Guide